



University of
Staffordshire

Group Project: Soapomorph.

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Course: Artificial Intelligence and Robotics

Staffordshire University

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Techniques (potentials)

First stage:

- Haptic pen – One possible use to try to make the main parts of the model.
- 3D Scanning – we have a skull model to scan to put into the main model.
- Design on Blender – Another possible use to try to make the main parts of the model.

Second stage:

- 3D printing – As we decided to go with the main part of our material being PLA 2.86.
- Silicone – For the inner jaw to keep it sterile/clean easier as plastic can have gaps that build bacteria. This is the most direct contact for users on the model.

Third Stage:

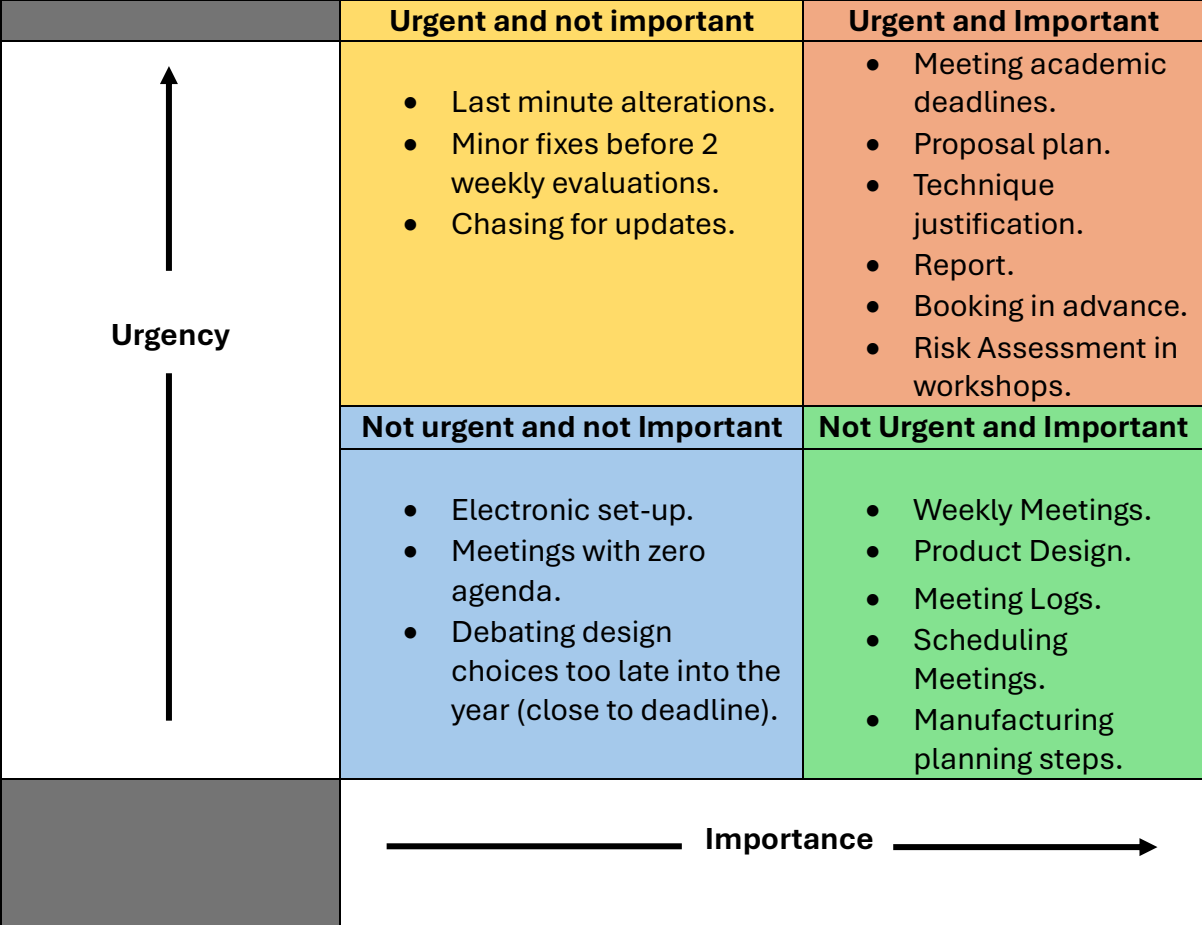
- Testing and evaluation.

MOSCOW

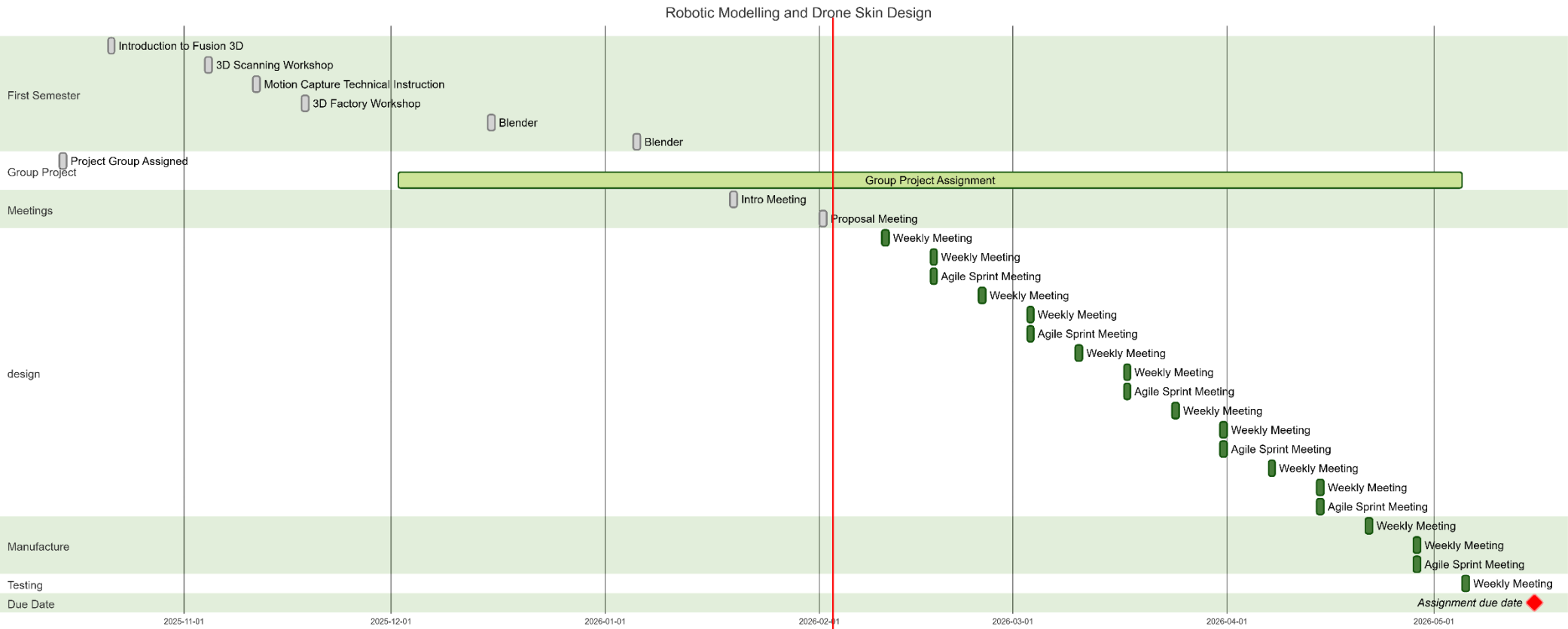
Must have	• Clear justification for using techniques during production. • Our own written report covering all techniques even ones we have not used. • Design model in report must have some form of planned movement.
Should have	• A physical product that shows some form of movement. • A budget plan (to be reviewed next meeting). • Evidence of safe and correct workshop practice adhering to the health and safety standards at Staffordshire university. (Risk assessment). • Testing and evaluation. • Design proof / diagramming/ program.
Could have	• Electronics to simulate the movement and perform visual demonstration. (This however is pushed back to the last stage as it is not important).

Will not have	- Any copyright material. - Advanced control systems, this is a prototype to showcase autonomous movement, and we will make use of the time we have.
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Eisenhower Matrix

 Urgency	Urgent and not important	Urgent and Important
	- Last minute alterations. - Minor fixes before 2 weekly evaluations. - Chasing for updates.	- Meeting academic deadlines. - Proposal plan. - Technique justification. - Report. - Booking in advance. - Risk Assessment in workshops.
	Not urgent and not Important	Not Urgent and Important
	- Electronic set-up. - Meetings with zero agenda. - Debating design choices too late into the year (close to deadline).	- Weekly Meetings. - Product Design. - Meeting Logs. - Scheduling Meetings. - Manufacturing planning steps.
	Importance	

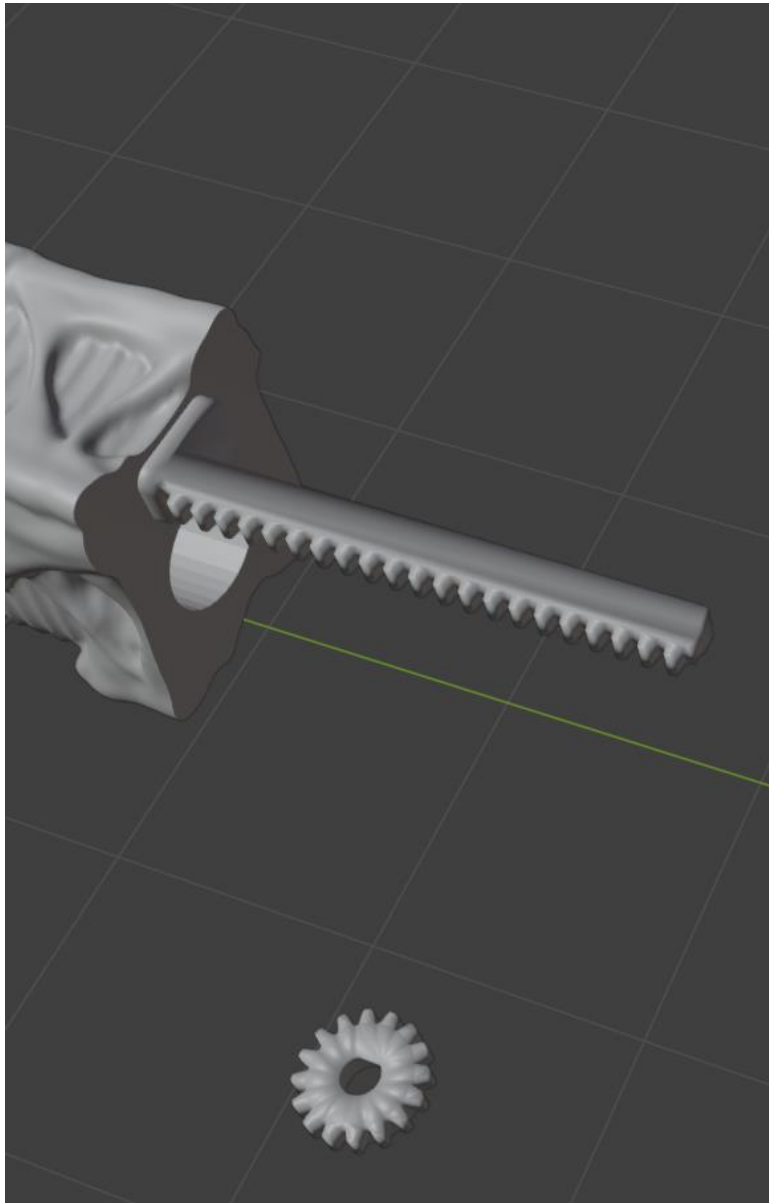
Gantt Chart Schedule



Design

Movement

The inner jaw will be the main source of simulated movement (in and out). Below is one idea a team member had to achieve this attached to a 360 servo.

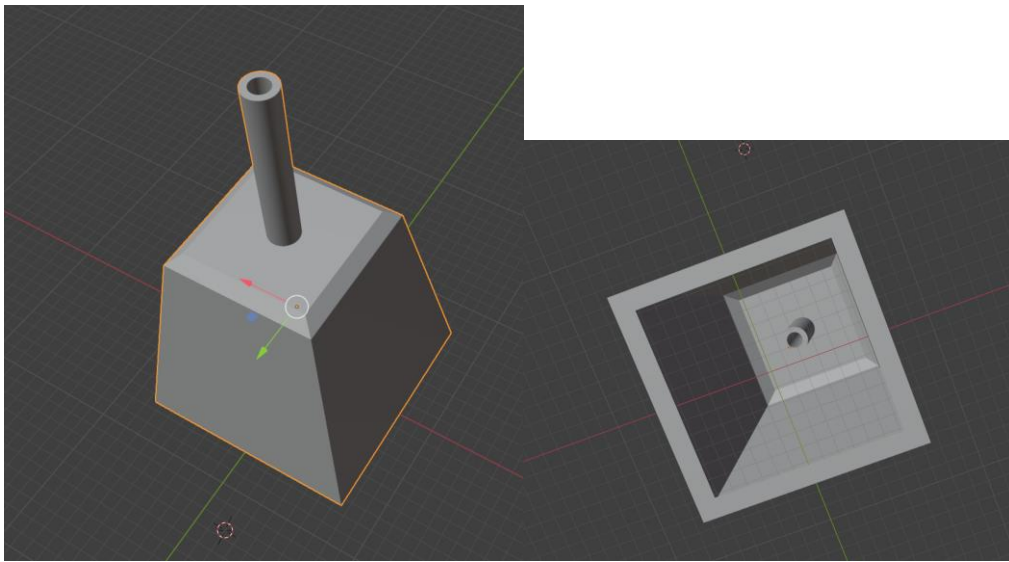


Mounting

The model will be between 25cm – 30cm in height.

Stand-alone base that is weighted so that when the soap gets low it won't fall over or become unbalanced.

Store bought soap insert easy to change and easy access, with an inner tube that runs up the model and into the inner jaw mouth.

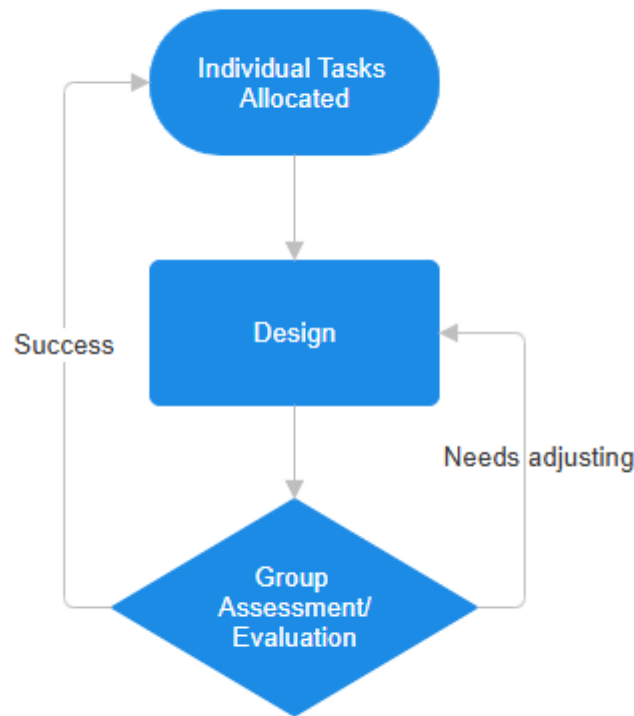


Above pictures are potentially what we will start with however we will revise (made briefly by a member of the group). We will be experimenting with different ideas on bases and weights until we have decided on a base we like.

We have also discussed the use of a wall mounted model however, for the purpose of a prototype we decided to go for a handheld model that's easily moved.

Agile Methodology Practice

We will be using the agile methodology production method to keep in constant communication with each other, doing 2-week sprints where a goal is set per person, and we will evaluate and assess changes or the next steps going forward. This allows time for mistakes and a flexible teamwork environment.



Electronics

We plan to provide our own electronics we have at home to cut down costs on the assumption that we will get any donated electronics back after the work has been graded.

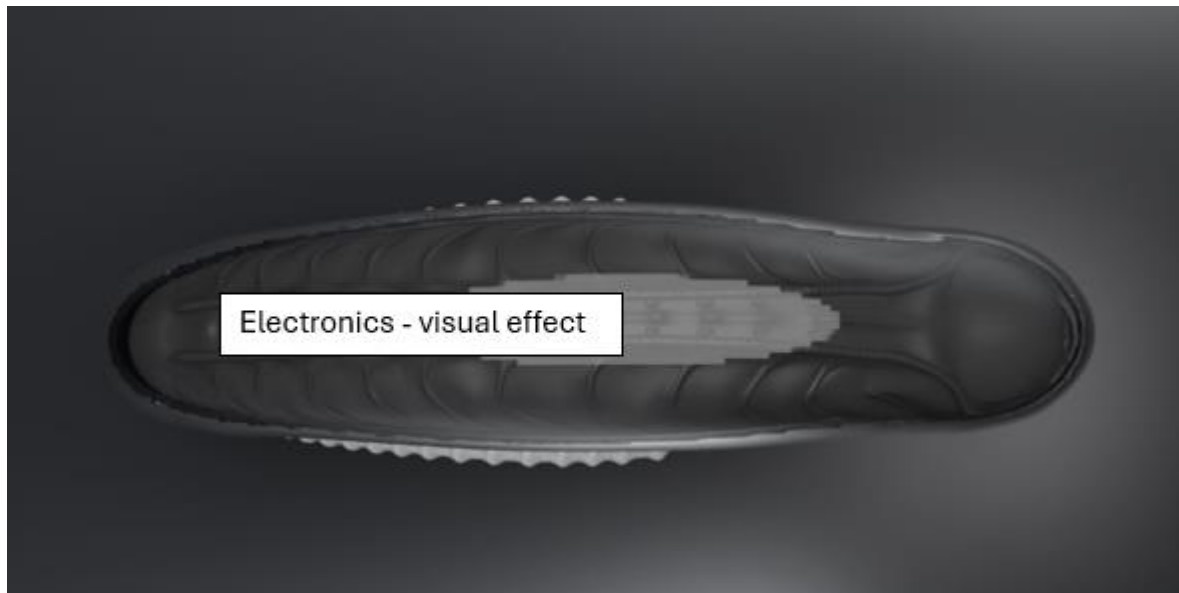
Not the focus of the project as we will strictly be focusing on design and movement, however a bonus should we be able to accomplish it later. We plan to put an ultrasonic sensor as the main point of contact to initiate the motion of the inner jaw going forward and the peristaltic pump to activate to dispense the soap.

An Arduino nano/ uno for a microcontroller to program specific tasks upon triggers.

LED system to show when the soap is running low with a float switch.

A battery pack easily accessible with AA inserts to allow convenience as most bathrooms in the UK do not provide a socket or power source. This also allows for us to keep costs down on a prototype.

A transparent top to put the electronics into for a visual effect and to keep it above liquid in case of risk of leaks.



References:

- <https://mermaid.ai/> - Creating Gantt Chart – Last used 01/02/2026.
- <https://www.smartdraw.com/flowchart/flowchart-maker.htm> - Used for Agile Methodology Flowchart – Last used 02/02/2026.
- <https://www.cgtrader.com/3d-print-models/miniatures/figurines/xenomorph-head-scale-model> - Only used to demonstrate electronics location not for our own work – Last used 02/02/2026.